

Quick Activity 1: School Supplies Calculator

A notebook costs ₱45, and a pen cost ₱12. A student buys several notebooks and pens and pays a certain amount. Develop an algorithm that calculates the **total cost** and **change**.

Input	Process	Output
Quantity of notebook, Quantity of pens, payment	Calculate notebook cost, calculate pen cost, calculate total cost, calculate change	Total cost, change

Descomposition

1. Input the Quantity of notebook.
2. Input the Quantity of pen.
3. Input the payment amount.
4. Calculate the notebook cost.
5. Calculate the pen cost.
6. Calculate the total cost.
7. Calculate the change.
8. Display the total cost and change.

Pseudocode

START

 READ notebookPrice, penPrice, paymentAmount

 notebookCost $\leftarrow$ quantity x 45 penCost $\leftarrow$

 quantity x 12 totalCost $\leftarrow$ notebookCost +

 penCost change $\leftarrow$ payment – totalCost

 DISPLAY totalCost, change

END

Dry Run Calculations

Test Case 1 Input:

- Notebook: 3
- Pen = 2
- Payments = ₱200

Step	Variable	Calculation	Solution
1	Notebook Cost	3 x ₱45	₱135
2	Pen Cost	2 x ₱12	₱24
3	Total Cost	135 + ₱ 24	₱159
4	Change	200 - ₱159	₱41

Output

Total Cost = ₱159

Change = ₱41

Test Case 2

Input:

- Notebook = 2
- Pen = 3
- Payment = ₱150

Step	Variable	Calculation	Solution
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1	Notebook Cost	2 x ₱45	₱90
2	Pen Cost	3 x ₱12	₱36
3	Total Cost	90 + ₱36	₱126
4	Change	150 - ₱126	₱24

Output

Total Cost = ₱126

Change = ₱24